

SHOFIA FAUZIYAH

Course by NordEdu
SQL #2

15 DAYS OF USING
SQL

Mastering Data Retrieval & Management with
PostgreSQL

Challenge #1

Used to select and return data

SYNTAX

```
Query  Query History
1  -- How many payment were made by the customer with customer_id = 100?
2  SELECT COUNT (*)
3  FROM payment
4  WHERE customer_id = 100;
```

OUTPUT

Data Output	
	count
1	24

notes

Menghitung jumlah transaksi pembayaran yang dimiliki oleh pelanggan dengan customer_id = 100, hasil execute menghitung sebanyak 24 transaksi.

Challenge #2

SYNTAX

```
6  -- What is the last name of our customer with first name 'ERICA'?
7  SELECT
8      first_name,
9      last_name
10 FROM customer
11 WHERE first_name = 'ERICA';
```

OUTPUT

Data Output			Messages	Notific
≡+	📄	▼	📋	▼
	first_name text	last_name text	🔒	🔒
1	ERICA	MATTHEWS		

cek >>

```
12 -- What is the last name of
13 -- our customer with first name 'ERICA'?
14 SELECT COUNT (*)
15 FROM customer
16 WHERE first_name = 'ERICA';
```

Data Output		M
≡+	📄	▼
	count bigint	🔒
1	1	

notes

Hanya menampilkan daftar customer yang nama depannya ERICA dan nama belakangnya. Untuk menghindari 'case sensitive' dan memastikan hasil sesuai bisa menggunakan COUNT seperti challenge sebelumnya. Dimana hasil menghitung jumlah yang memiliki nama ERIKA hanya ada 1.

Challenge #3

SYNTAX

```
17 -- The inventory manager asks you how rentals have not been returned yet (return_date is null).  
18 SELECT COUNT (*)  
19 FROM rental  
20 WHERE return_date IS NULL;
```

OUTPUT

Data Output	
	
	count bigint
1	183

notes

Menghitung jumlah data penyewaan yang belum dikembalikan, dengan cara menghitung jumlah kolom return_date yang masih kosong atau (NULL).

Challenge #4

penjelasan lebih lanjut >>

SYNTAX

```
22 -- The sales manager asks you how for a list of all the
23 -- payment_ids with an amount less than or equal to $2.
24 -- Include payment_id and the amount.
25 SELECT
26     payment_id,
27     amount
28 FROM payment
29 WHERE amount <= 2;
```

OUTPUT

Data Output			Messages	Notificati
≡+ ▼ ▼				
	payment_id integer	amount numeric (5,2)		
1	16050	1.99		
2	16051	0.99		
3	16053	0.99		
4	16056	1.99		
5	16059	0.99		
6	16062	0.99		
7	16064	0.99		
8	16077	0.99		

notes

Menampilkan payment_id dan amount dari tabel payment untuk pembayaran dengan nilai kurang dari atau sama dengan \$2
terdapat Total rows: 3644 dengan pembayaran

Challenge #4 lanjutan

```
22 -- The sales manager asks you how for a list of all the
23 -- payment_ids with an amount less than or equal to $2.
24 -- Include payment_id and the amount.
25 SELECT
26     payment_id,
27     amount
28 FROM payment
29 WHERE amount >= 2
30 ORDER BY amount ASC;
```

Data Output Messages Notifications

	payment_id integer	amount numeric (5,2)
1	22209	2.99

```
22 -- The sales manager asks you how for a list of all the
23 -- payment_ids with an amount less than or equal to $2.
24 -- Include payment_id and the amount.
25 SELECT
26     payment_id,
27     amount
28 FROM payment
29 WHERE amount = 2;
```

Data Output Messages Notifications

	payment_id integer	amount numeric (5,2)
--	-----------------------	-------------------------

notes

Pada challenge 4 nilai amount yang paling mendekati 2 adalah 1.99. untuk memastikan bisa menggunakan kondisi (where=2) dimana memang tidak ada yang bernilai amount=2. Bisa dipastikan juga dengan melihat nilai (amount>=2) dengan kondisi ASC agar terlihat urutan dari nilai yang paling mendekati 2.

Challenge #5

notes

SYNTAX

```
32 -- The suppcity manager asks you about a list of all the payment of
33 -- the customer 322, 346 and 354 where the amount is either less
34 -- than $2 or greater than $10.
35 SELECT *
36 FROM payment
37 WHERE customer_id IN (322,346,354)
38 AND (amount <=2 OR amount > 10)
39 ORDER BY
40     customer_id ASC,
41     amount DESC;
```

OUTPUT

Data Output							Messages	Notifications
≡+ 📄 ▼ 📋 ▼ 🗑 🗄 ⬇ 📈 SQL Showing rows: 1 to 25 ✎ 📄 P								
	payment_id integer	customer_id smallint	staff_id smallint	rental_id integer	amount numeric (5,2)	payment_date timestamp with time zone		
1	25784	322	2	3627	1.99	2020-04-06 10:47:51.996577+...		
2	25794	322	1	9252	1.99	2020-04-30 18:48:25.996577+...		
3	16167	322	2	166	0.99	2020-01-26 08:17:37.996577+...		

Menampilkan semua data pembayaran dari customer 322, 346, dan 354
Kondisi pertama dengan nilai kurang dari atau sama dengan \$2 atau lebih dari \$10, lalu mengurutkannya berdasarkan customer_id dari nilai terkecil dan amount berdasarkan nilai terbesar.

Challenge #6

SYNTAX

```
43 -- There have been 6 complaints of customers about their
44 -- payments.
45 -- customer_id: 12,25,67,93,124,234
46 -- The concerned payments are all the payments of these
47 -- customers with amounts 4.99, 7.99 and 9.99 in January 2020.
48 SELECT *
49 FROM payment
50 WHERE customer_id IN (12,25,67,93,124,234)
51 AND amount IN (7.99,4.99,9.99)
52 AND payment_date BETWEEN '2020-01-01'AND '2020-02-01';
```

OUTPUT

Data Output Messages Notifications							
Showing rows: 1 to 7							
	payment_id integer	customer_id smallint	staff_id smallint	rental_id integer	amount numeric (5,2)	payment_date timestamp with time zone	
1	16699	12	1	988	4.99	2020-01-31 04:36:29.996577+...	
2	16700	12	1	1084	4.99	2020-01-31 16:38:43.996577+...	
3	16736	25	1	90	7.99	2020-01-25 19:59:51.996577+...	
4	16822	67	2	331	9.99	2020-01-27 07:50:52.996577+...	
5	16870	93	1	1025	4.99	2020-01-31 09:10:03.996577+...	
6	16929	124	2	1039	4.99	2020-01-31 11:00:55.996577+...	
7	17116	234	2	1125	4.99	2020-01-31 22:52:10.996577+...	

notes

Menampilkan semua data pembayaran dari customer 12, 25, 67, 93, 124, dan 234 dengan nominal 4.99, 7.99, atau 9.99, yang dilakukan pada periode 1 Januari 2020 hingga 1 Februari 2020. Terdapat 3 kondisi berdasarkan :

- customer_id
- amount
- payment_date

pada case ini batas filter date terdapat batasan waktu, sehingga batas ahir bulan januari menggunakan 01 februari.

Challenge #7

SYNTAX

```
54 -- There have been some faulty payments and you need to help to
55 -- found out how many payments have been affected.
56 -- How many payments have been made on January 26th and 27th
57 -- 2020 (including entire 27th) with an amount between 1.99 and 3.99?
58 SELECT COUNT(*)
59 FROM payment
60 WHERE payment_date >= '2020-01-26'
61 AND payment_date <= '2020-01-27'
62 AND amount BETWEEN 1.99 AND 3.99;
```

OUTPUT

	count bigint
1	53

notes

Menghitung jumlah pembayaran yang dilakukan pada 26–27 Januari 2020 dengan nominal antara 1.99 hingga 3.99 dari tabel payment, pastikan untuk batas tanggal dengan waktunya.

Challenge #8

SYNTAX

```
64 -- You need to help the inventory manager to find out:
65 -- How many movies are there that contain the "Documentary" in
66 -- the description?
67 SELECT COUNT(*)
68 FROM film
69 WHERE description LIKE '%Documentary%';
```

OUTPUT

	count bigint 
1	101

notes

Terdapat 101 film yang memiliki deskripsi "Documentay" di dalamnya.

Challenge #9

SYNTAX

```
72 -- How many customers are there with a first name that is
73 -- 3 letters long and either an 'X' or a 'Y' as the last letter in the last
74 -- name?
75 SELECT COUNT(*)
76 FROM customer
77 WHERE first_name LIKE '___'
78 AND (
79     last_name LIKE '%X'
80     OR last_name LIKE '%Y');
```

OUTPUT

	count bigint
1	3

```
82 SELECT *
83 -- COUNT(*)
84 FROM customer
85 WHERE first_name LIKE '___'
86 AND (
87     last_name LIKE '%X')
88     -- last_name LIKE '%Y')
89 ;
```

Data OutputMessagesNotifications

	customer_id [PK] integer	store_id smallint	first_name text	last_name text
1	538	2	TED	BREAUX

```
82 SELECT *
83 -- COUNT(*)
84 FROM customer
85 WHERE first_name LIKE '___'
86 AND (
87     -- last_name LIKE '%X')
88     last_name LIKE '%Y')
89 ;
```

Data OutputMessagesNotifications

	customer_id [PK] integer	store_id smallint	first_name text	last_name text
1	181	2	ANA	BRADLEY
2	468	1	TIM	CARY

notes

Terdapat 3 customer dengan kondisi yang memiliki banyaknya 3 huruf pada nama depan dan nama belakang berakhiran huruf X atau Y.

hasil tes menunjukkan ada satu customer dengan ahiran huruf X pada nama belakang yaitu:

"TED" "BREAUX"

dan hasil tel untuk nama belakang dengan ahiran huruf Y yaitu:

"ANA" "BRADLEY"

"TIM" "CARY"

Challenge For Today
#DAY2

Challenge #1

SYNTAX

```
98 -- How many movies are there that contain 'Saga'
99 -- in the description and where the title starts either
100 -- with 'A' or ends with 'R'?
101 -- Use the alias 'no_of_movies'.
102 SELECT COUNT(*) AS no_of_movies
103 FROM film
104 WHERE description LIKE '%Saga%'
105 AND (title LIKE 'A%' OR title LIKE '%R');
```

OUTPUT

Data Output		Message
	no_of_movies bigint	
1		14

notes

Sebanyak 14 film yang memiliki deskripsi atau mengandung kata "Saga" dan judulnya diawali huruf A atau diakhiri huruf R.

Challenge #2

notes

SYNTAX

```
107 -- Create a list of all customers where the first name contains
108 -- 'ER' and has an 'A' as the second letter.
109 -- Order the results by the last name descendingly.
110 SELECT *
111 FROM customer
112 WHERE first_name LIKE '%ER%'
113 AND first_name LIKE '_A%'
114 ORDER BY last_name DESC;
```

Terdapat 5 customer yang memiliki kondisi nama depannya mengandung huruf "ER" dan memiliki huruf A sebagai huruf kedua, lalu mengurutkannya berdasarkan last_name secara menurun (DESC)

OUTPUT

Data Output Messages Notifications										
Showing rows: 1 to 5 Page No: 1 of 1										
	customer_id [PK] integer	store_id smallint	first_name text	last_name text	email text	address_id smallint	activebool boolean	create_date date	last_update timestamp wi	
1	61	2	KATHERINE	RIVERA	KATHERINE.RIVERA@sakilacustomer.org	65	true	2020-02-14	2020-02-15 1	
2	339	2	WALTER	PERRYMAN	WALTER.PERRYMAN@sakilacustomer.org	344	true	2020-02-14	2020-02-15 1	
3	535	1	JAVIER	ELROD	JAVIER.ELROD@sakilacustomer.org	541	true	2020-02-14	2020-02-15 1	
4	46	2	CATHERINE	CAMPBELL	CATHERINE.CAMPBELL@sakilacustomer....	50	true	2020-02-14	2020-02-15 1	
5	149	1	VALERIE	BLACK	VALERIE.BLACK@sakilacustomer.org	153	true	2020-02-14	2020-02-15 1	

Challenge #8

SYNTAX

```
116 -- How many payments are there where the amount is either 0
117 -- or is between 3.99 and 7.99 and in the same time has
118 -- happened on 2020-05-01.
119 SELECT COUNT (*)
120 FROM payment
121 WHERE payment_date BETWEEN '2020-05-01 00:00:00'
122                        AND '2020-05-01 23:59:59'
123 AND (amount = 0 OR amount BETWEEN 3.99 AND 7.99);
```

```
126 SELECT amount, COUNT(*) AS jumlah
127 FROM payment
128 WHERE payment_date
129 BETWEEN '2020-05-01 00:00:00'
130 AND '2020-05-01 23:59:59'
131 AND (amount = 0 OR amount BETWEEN 3.99 AND 7.99)
132 GROUP BY amount
133 ORDER BY amount;
```

OUTPUT

Data Output	
≡+	📄
	count bigint 🔒
1	160

notes

Banyaknya payment dalam yang terjadi pada 1 Mei 2020, dengan kondisi dimana nilai pembayaran 0 atau berada di antara 3.99 hingga 7.99 sebanyak 160.
Pada gambar sebelah kanan untuk memastikan jumlah payment berdasarkan nilai amount nya lebih detail yang jika dijumlah hasilnya sama 160.

Data Output			Messages	Noti
≡+	📄	▼	📄	▼
	amount numeric (5,2) 🔒	jumlah bigint 🔒		
1	3.99	23		
2	4.99	70		
3	5.99	27		
4	6.99	19		
5	7.99	21		

THANKYOU